

Individual differences in the computational processes that support face recognition

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1. Introduction

Being able to quickly and accurately recognise other people is an important aspect of everyday life, which impacts social interactions as well as more specific tasks, such as eye-witness identification and passport control checks. However, face recognition ability varies across the population. For instance, people with developmental prosopagnosia show relatively poor face recognition ability (Dobel et al., 2007; McConachie, 1976), whereas so-called “super-recognisers” show a remarkably high face recognition ability compared to the average person (Bobak et al., 2016; Russell et al., 2009). Even if we ignore these two extreme cases, face recognition ability varies throughout the general population, especially when identifying unfamiliar individuals (Dowsett & Burton, 2015; White et al., 2014). Unsurprisingly, therefore, recent research has attempted to understand the mental processes that determine variation in face recognition ability as a route to uncover the architecture of a key dimension of social perception and cognition [REF??]. Here, in this project, we move this research field forward by combining approaches from experimental, differential and mathematical psychology, to investigate how individual differences in face recognition ability relate to the computational processes that support face recognition.

Attempts to understand differences between individuals has a long history in psychological research. Indeed, differential psychology represents a major branch of psychological research that aims to understand individual differences in thought, behaviour and ability (Lubinski, 2000; Marsh & Boag, 2013). In the domain of face recognition, there is a growing interest in investigating individual differences across the entire spectrum of face recognition ability, rather than focusing on specific subgroups, such as “super-recognisers” (White & Kitchen, 2022). To this end, several assessment tools have been designed to measure different facets of face recognition performance (Duchaine & Nakayama (2006); Burton et al. (2010); Russell et al. (2009)). For example, the Cambridge Face Memory Test (Duchaine & Nakayama (2006)) measures how many previously seen faces individuals can accurately recognise, with performance varying across the general population and being poorer in individuals with developmental prosopagnosia. As such, these tools appear to have sufficient validity to represent a marker of an individuals face recognition ability.

Using these tools as a marker of face recognition ability, past research has estimated the association between the performance on these tests and broader perceptual and cognitive processes (McCaffery et al., 2018; Noyes et al., 2018). For instance, one line of research has studied the extent to which individual difference in face processing reflect domain-general or more domain-specific differences in perceptual and cognitive abilities (McCaffery et al., 2018; Verhallen et al., 2017). Some studies have shown that face and general object perception ability are positively, but modestly associated (Dennett et al., 2012a; Richler et al., n.d.; Wilmer et al., 2012). However, associations between performance in face recognition tasks and non-face recognition tasks remain constantly weaker than associations between different face recognition tasks (Fysh et al., 2020; McCaffery et al., 2018; Wilmer et al., 2010). This could be considered as evidence that some part of the variance between face and object recognition tasks performance is explained by face-specific processing (Richler et al., 2019).

The use of directly observed or manifest measures, such as reaction time and accuracy, has made a valuable contribution to understanding the relationship between psychological processes in general and in the context of face perception. However, performance on such tests remains a distant proxy of the underlying cognitive processes at work during face recognition, which makes them far from ideal for capturing individual differences in cognitive processes (Haines et al., 2020). An alternative approach to this problem has emerged in the domain of mathematical psychology and involves using a computational model to estimate latent computational parameters, which can be closer to the cognitive processes of interest (Yarkoni, 2022). For instance, evidence-accumulation models translate the distribution of response times for correct and incorrect responses into parameters reflecting latent computational processes that are supposed to underlie the decision-making process (see Ratcliff et al., 2016 for a review). The linear ballistic accumulator (LBA) model (?@fig-lba) belongs to the set of such evidence-accumulation models (brownSimplestComplete-Model). In two-alternatives forced choice tasks, this model describes the accumulation of evidence (drift rate parameter) towards the two alternative options. Once the accumulation of evidence crosses the threshold, the corresponding response is then chosen. Moreover, these parameters can

be interpreted in psychologically meaningful ways: the drift rate parameter reflect the signal-to-noise ratio provided by the stimulus and the threshold parameter reflects the response caution towards a response (the amount of evidence needed to trigger a particular response; Myers et al., 2022). From an individual differences perspective, one can then study how variation in these latent computational parameters relates to key aspects of social behaviour, such as face recognition ability (White et al., 2016).

Approaches from differential psychology have historically run alongside, and largely separate to, research traditions from experimental psychology, which aim to estimate the average effect (aggregated across individuals) of an experimental manipulation (Myers & Hansen, 2006). For example, we previously used an experimental approach in combination with evidence accumulation modelling to study how familiarity impacts the computational processes that support face recognition [add ref: fournier2026familiarity]. We found that different types of familiarity (visually familiar faces vs. famous faces) had a similar impact on the computational processes involved in the decision-making process of face recognition. Compared to learned faces, novel faces induced a greater accumulation of evidence (drift rate), as well as modest increase in response caution (threshold). This multifaceted effect of familiarity on the face recognition process would not have been possible using a traditional separate analysis of response times and accuracy. Although we showed that these results could be replicated in both laboratory and online settings, the work aimed to estimate an average effect across a relatively small number of individuals ($N \sim 30$), rather than investigate individual differences. As such, (add ref: fournier2026familiarity) were unable to estimate how variation in face recognition ability between individuals shapes the computational processes that determine how face recognition judgements are made.

In the present study, therefore, we combined approaches from differential, experimental and mathematical psychology to investigate how latent computational parameters in face recognition relate to face recognition ability. More specifically, based on prior results from our lab [add ref: fournier2026familiarity], we investigated the extent to which drift rate and threshold vary as a function of face recognition ability specifically or of a more general object recognition ability. This approach can provide insights regarding the domain-specificity of the computational processes involved in face recognition. To do so, we used a similar face memory task as used previously by

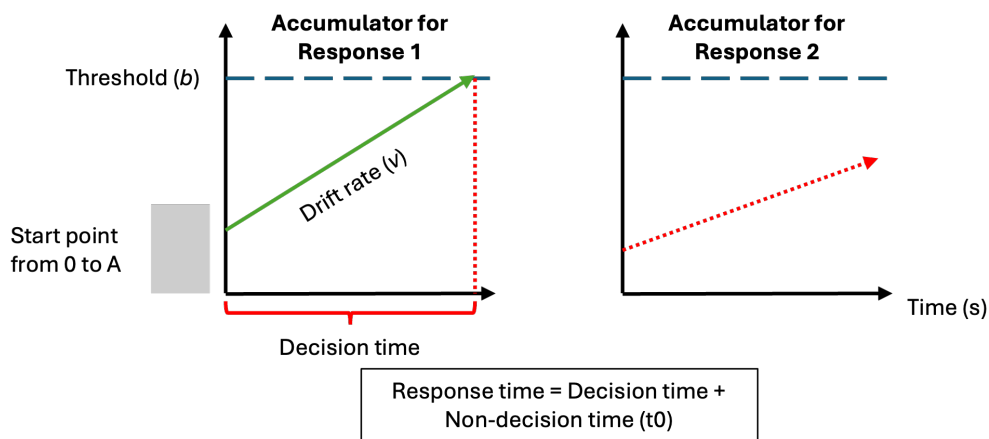


Figure 1. Schematic of the linear ballistic accumulator (LBA) model

This diagram represents the accumulation process one trial of a perceptual decision task for the linear ballistic accumulator (LBA; Brown & Heathcote (2008)). Participants are required to choose between two options (Response 1 and Response 2). In a LBA model, there is a separated accumulation of evidence (v) for each response option. In this example, the correct response is Response 1. The accumulator on the left is the accumulator for the correct response (Response 1), and the green line in represents the rate at which the evidence accumulates in favor of that response option. The accumulator on the right is the accumulator for the incorrect response (Response 2), and the dash red line represents the rate at which the evidence accumulates in favor of this response option. Once the accumulated evidence crosses the threshold (b) of a response option, the corresponding response is chosen (in this case Response 1).

(add ref: [fournier2026familiarity](#)) to compute key latent parameters, whilst also measuring face recognition ability ([Duchaine & Nakayama, 2006](#)) and general object recognition ability ([Dennett et al., 2012a](#)) in a considerably larger sample of participants ($N \sim 300$) than our prior work to enable more robust estimates of between participant individual differences.

Based on past research, we might expect that drift rate (the quality of the evidence accumulation) to positively correlate with face recognition ability, but it is less clear how threshold (response caution towards a response) may vary. For instance, a better face recognition could result in being more cautious or less cautious. Moreover, how these patterns vary when accounting for familiarity effect (i.e. learned face vs. novel faces) remains unknown.

2. Method and Materials

2.1. Pre-registration and open science

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study ([Simons et al., 2017](#)). The research question, hypotheses, experimental design, planned analysis, and exclusion criteria were preregistered (<https://osf.io/b74gt/overview>). Raw data and analysis code are available on Github ([link github](#)).

2.2. Participants

All participant were recruited on the Prolific platform (<https://www.prolific.com>). Participants had to be located in the United Kingdom or in the United States, be aged between 18-65 years old, with normal or corrected-to-normal vision and speak fluent English. Furthermore, Prolific offers the option to chose the approval rate of the participants willing to take part in the study. The approval rate is the percentage for which participants has been approved in previous experiment. An approval rating between 95% and 100% was chosen in our study to try to ensure high data quality. 383 participants completed the experiment. Following our preregistered exclusion criterion, 2 participants were excluded because their percentage of missed trials was higher than 40%, 3 because their percentage of RT outside the established range (0.2 to 2 seconds) were higher than 25% and 78 were removed because the overall accuracy was lower than 55%. In the end, data from 300 participants (141 men and 159 women) were retained for the analysis and the LBA modeling. Ages ranged from 19 to 64 ($M_{age} = 42$, $SD_{age} = 11.2$). Following Prolific payment principles, each participant received 9£ for the completion of the study. This experiment was accepted by the ethics committee of the Federal Institute of technology (ETH) Zurich.

2.3. Material

Participant were asked to complete the study using a desktop computer. Beforehand, the paradigm was tested on three different operative systems (PC, Macintosh and Linux) and on three web browsers (Google Chrome, Safari and Firefox). Participant were aware that using any other combination of operative systems and web browsers might lead to the inability to complete the experiment and were asked to follow these directives to ensure a consistent procedure and participant experience.

2.4. Procedure

The experiment was divided into three experimental tasks. A face recognition task was used to estimate the LBA parameters from manifest measures (accuracy and response times). Face and car recognition ability were assessed using the Cambridge Face Memory Test and the Cambridge Car Memory Test. Participants always first underwent the face recognition task and the order of two memory tests was counterbalanced.

2.4.1. Face recognition task

The face recognition task was divided into two distinct parts: a familiarization part and a recognition part. During the familiarization part, unknown identities were presented on the screen and participants were instructed to memorize the identity of the faces. One familiarization trial consisted of the presentation of 6 face images, one after the other, showing the same identity but varying in viewing points (3/4 left view, frontal view and 3/4 right view) and the colorimetry

(colorful image, gray scale image). The duration of each image presentation was set to 0.75 second and each participant underwent 30 familiarization trials (for 30 unknown identities). During the familiarization, we introduced catch trials to ensure participants attention. After some familiarization trials, a gray scale image of a person's face was presented on the screen. Participants were asked to press the "q" key if the face belonged to the last seen identity or the "p" key if it was not. The faces seen during the familiarization were considered the "Learned faces" within-subject condition.

In the recognition part, participants completed three blocks of 90 recognition trials. Each block was separated from another by a short break. Half of the trials were learned identities trials and the other half were new, previously unseen identities trials. In each trial, a gray scale image of a face was shown on the screen. Participants were asked to choose if they had previously seen this face in the familiarization part or not. If they did, they should press the "q" key and if they did not, the "p" key. Participants were asked to respond as quickly and accurately as possible. It is important to note that participants never saw two pictures of the same learned identity in the same block and never saw the same picture (3/4 left view, frontal view, or 3/4 right view) of a learned identity during the entire recognition part. Furthermore, each novel identity was only presented once in the entire recognition part. Response time [s] and accuracy were collected after each recognition trial. A response time limit of 2 seconds has been set for each recognition trial. If participants took longer than 2 seconds to press a response key, the text "Too late!" appeared in red on the screen and the trial was discarded, but not replaced.

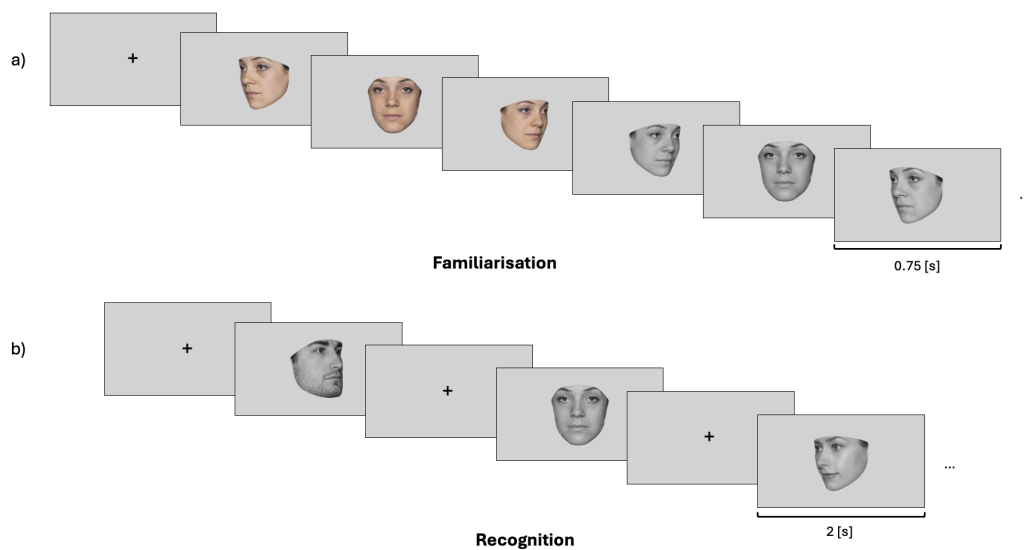


Figure 2. Experimental procedure of the face recognition task

The experimental procedure was divided into two parts: a familiarization part and a recognition part. a) In the familiarization part, participants were asked to memorize the face identity of 30 identities. For each identity, participants looked at 6 pictures depicting the same identity with different face orientation (3/4 left view, frontal view and 3/4 right view) and different colorimetry (color and gray scale images) in the order presented here: 3/4 left view, frontal view and 3/4 right view in color, followed by 3/4 right view, frontal view and 3/4 left view in gray scale. b) In each trial of the recognition part, participants had to decide if they recognize the face shown on the screen from the familiarization part. If they did, they had to press the "q" key and if they did not, the "p" key. They completed three blocks of 60 trials (half of them were from the familiarization part, half were novel faces). For the recognition part, only gray scale images of faces were used (for both learned faces and novel faces).

2.4.2. Cambridge Memory Tests

The method of the CFMT was described in [Duchaine & Nakayama \(2006\)](#). Participants learned 6 Caucasian male faces in three views and had to discriminate them later from distractor faces (no time response limit, three-alternative forced-choice task, Figure 3 a). Regarding the CCMT, the method has been described in [Dennett et al. \(2012a\)](#) and has a highly similar structure, format and scoring as the CFMT, but uses cars instead of faces as stimuli.

2.5. Stimuli

The face images used in the faces recognition task consisted of images selected from two different data sets of face images: the Face Research Lab London Set ([DeBruine, Lisa; Jones, Benedict \(2017\): <https://doi.org/10.6084/m9.figshare.5047666.v5>](#)) and the Color FERET Database (<https://www.nist.gov/itl/products-and-services/color-feret-database>). Each of these data sets is composed of images of different identities faces taken from different angle points. 102 identities were selected from the Face Research Lab London Set and 51 from the Color FERET Database. For each selected identity, three pictures of the face at three different view points (3/4 left view, frontal view and 3/4 right view) were cropped following the outline of the face.

To ensure that the stimuli that we used are reproducible, we used the webmorphR R Package ([DeBruine \(2022\)](#)) to crop the face images. The code used to crop the original pictures can be found on OSF ([OSF link](#)).

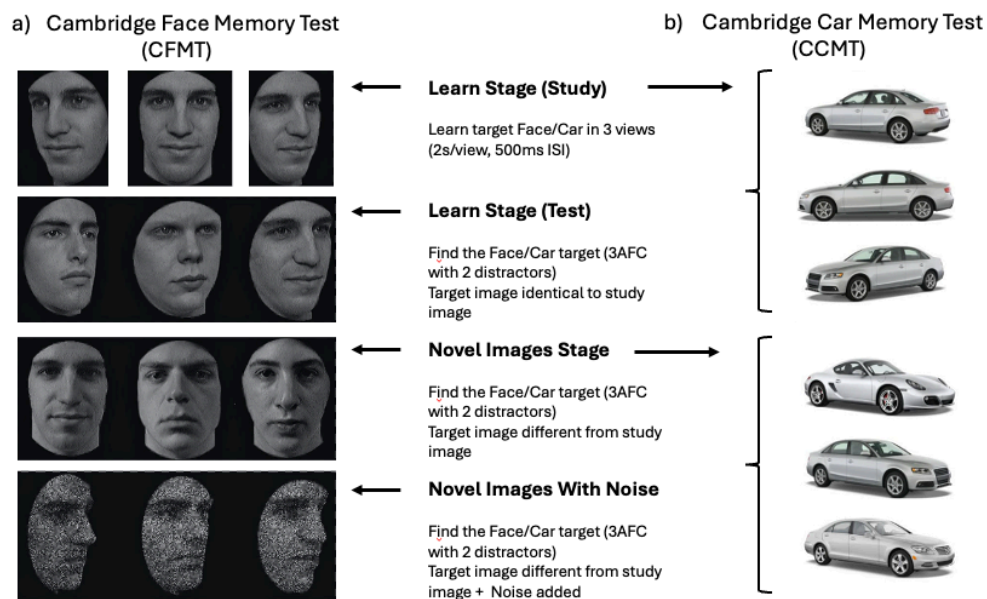


Figure 3. The two Cambridge Memory Tests

Illustration of (a) the Cambridge Face Memory Test (CFMT; ref Duchain & Nakayama, 2006) and (b) the Cambridge Car Memory Test (CCMT; red Dennett et al., 2012a). These two tests are highly similar, differing only in the category of the presented stimuli. Both tests are composed of three stages: Learn (including Study and Test), Novel Images and Novel Images with Noise. In the Learn stage, participants learn target faces/cars and are then tested on the recognition of the identical images to the learned targets. In the Novel Images and Novel Images with Noise stages, recognition targets are different pictures of the same learned targets, but differing in views, lightning or both. For the CCMT (b), the car pictures showed here are not the one used in the CCMT and served only as a representation of the stimuli used in the Learn stage and the Novel stages of the CCMT. This illustration has been taken and slightly modified from [Dennett et al. \(2012b\)](#). 3AFC = three-alternative forced choice; ISI = interstimulus interval.

2.6. Evidence accumulation modeling

The model and priors specification as well as the model estimation are all identical to the one described in (ref: Fournier et al., in press). The priors specifications are described in Supplementary Materials.

2.6.1. Model specification

A hierarchical LBA models was fit to the accuracy and response times of the face recognition task. The model specifications are outlined in our preregistration (Table 1). The LBA framework assumes a separate evidence accumulator for each response alternative, with potentially different parameter values. Accordingly, one accumulator corresponded to the “Learned” response and the other to the “Novel” response. Threshold was allowed to vary by participants’ response (Learned/Novel) and mean drift rate varied as a function of both stimulus type (Learned/Novel) and whether the accumulator corresponded to the correct response for that stimulus (TRUE vs. FALSE). For example, when a Learned stimulus was presented, the Learned accumulator was designed as TRUE, whereas the Novel accumulator was designed as FALSE. The difference in drift rate between the TRUE and FALSE accumulators reflects the quality of information supporting a decision, with larger differences indicating stronger and more discriminative evidence accumulation (Lewandowsky & Oberauer (2018)). The standard deviation of the TRUE accumulator was allowed to vary according to accumulator match status, whereas the standard deviation of the FALSE accumulator was constrained to be constant to ensure model identifiability (Donkin et al. (2009)). In the preregistered model, a single start-point parameter (A), and single non-decision time parameter (t0) were estimated. Alternative model specifications that permitted variation in these parameters are reported in the Supplementary Materials.

Table 1. Formula specification for the pre-registered linear ballistic accumulator model.

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	no	no	no	-
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	no	yes	no	-
Non-decision time	t0	no	no	no	-

Condition = training condition (Learned vs. Novel); Response = key-press response (same vs. different); Accuracy = (true/correct vs false/incorrect); SD = standard deviation.

2.6.2. Model estimation

Bayesian model estimated was carried using the EMC2 R package (Stevenson et al., 2026). Priors were chosen to be weakly-informative, meaning that they have little influence on the estimation (for more details regarding the priors of each model, see Supplementary Materials). Two sampling steps were used to estimate the parameters. First, a separate estimation of the fixed effects for each participant was conducted. Parameter estimations were then used as the starting point for the full hierarchical parameter estimation.

2.6.3. Model fit

To ensure that the fit of our model can capture most of the main trends in the data, graphical summaries of the posterior predictions were examined. Convergence plots and posterior predictive checks can be found in Supplementary Materials.

2.7. Data analysis

Regarding the general analytical strategy, two main approaches were considered. Correlations and data visualizations were used to characterize the zero-order relationships between parameters of interest (drift rate and threshold) from the LBA model with face and object recognition ability. Linear regression was then conducted to estimate how much LBA parameters vary as a function of

face recognition ability, as well as how much these parameters vary after adjusting for car recognition ability. The details of the analysis plan were preregistered (<https://osf.io/b74gt/overview>).

2.7.1. *LBA parameters*

In order to make inference about the magnitude of familiarity effects, differences in parameter estimates across conditions were compared. To summarize the distribution of parameters of interest, we report the median of the posterior distribution together with 95% quantile interval of the posterior distribution.

2.7.2. *Classical correlational analysis*

The median value of each parameter’s posterior distribution was correlated with the two centred scores of each of the Cambridge Memory Tests to assess the relationship between these variables.

2.7.3. *Bayesian plausible value correlational analysis*

In addition to the classical correlational analysis described above, we conducted a Bayesian plausible value correlation approach (Ly et al., 2017; Matzke et al., 2017). Instead of using point estimates (median value of each parameter’s posterior distribution), this approach permits the estimation of uncertainty in likely correlational values by correlating the parameter estimation from each sample (or “draw”) from the posterior distribution with the memory scores. A distribution of the plausible correlations between each parameter and the memory scores can then obtained. These distributions can compared between stimuli conditions (Learned vs Novel) or memory scores (face vs car) using difference contrasts. It allowed us to estimate if the strength of the correlation differs or not between conditions of interest.

2.7.4. *Linear regression*

Based on our pilot results, we preregistered two Bayesian linear regression models to quantify the extent to which drift rate vary as a function of face recognition ability alone and when accounting for car recognition ability. The results of the linear regresison for the threshold parameter are available in the Supplementary materials. An index coding approach was used (no grand intercept) where each condition of interest has its own intercept (see Table 2 for the details of each regression models).

Table 2. Formula specification for the pre-registered linear regression models.

Parameter	Condition	Ability	Formula
Median drift rate	familiarity:accuracy	Face	Median drift rate: 0 + familiarity:accuracy + face_ability + (0 participant)
Median drift rate	familiarity:accuracy	Face + Car	Median drift rate: 0 + familiarity:accuracy + face_ability + car_ability + (0 participant)

familiarity = face familiarity (Learned vs. Novel); accuracy = response accuracy (TRUE vs. FALSE)

2.7.5. *Inferential criteria*

Using a Bayesian approach in our regression analysis allowed us to evaluate the posterior distributions of the parameter of our predictor of interest (Face Memory score). To evaluate these posterior distributions, we used three complementary criteria: sign, size and precision (Cumming, 2014; Kruschke & Liddell, 2018; McElreath, n.d.). To address the sign (or direction) of the effect, we reported if the median value is positive or negative. Second, we evaluated the size or magnitude of the posterior distribution by considering the size of the median value as well as the extent to which lower or upper bounds of the 95% quantile interval approach zero (no relationship). For example, if the 95% quantile interval excluded zero, we would have been reasonably confident in our inference of a non-zero effect. Furthermore, we used the width of the 95% quantile intervals as an indicator of estimation precision. Whereas narrow estimates indicate precise estimates, wider intervals indicate substantial uncertainty about the estimated parameter value, which may warrant a more cautious interpretation.

3. Results

3.1. LBA parameters

In this section, we describe the posterior distributions of the drift rate and threshold parameters for both Learned and Novel faces estimated from the distribution of response times and accuracy collected in the face recognition task. We then contrasted these distributions to

Posterior distributions for drift rate and threshold parameters are visualized in Figure 4 and Figure 5 and the medians and 95% quantile intervals are summarised in Table 3.

Threshold: Inspection of the posterior distributions shows that there is an effect of familiarity on the threshold parameter. The threshold parameter is higher for the Novel response compared to the Learned response.

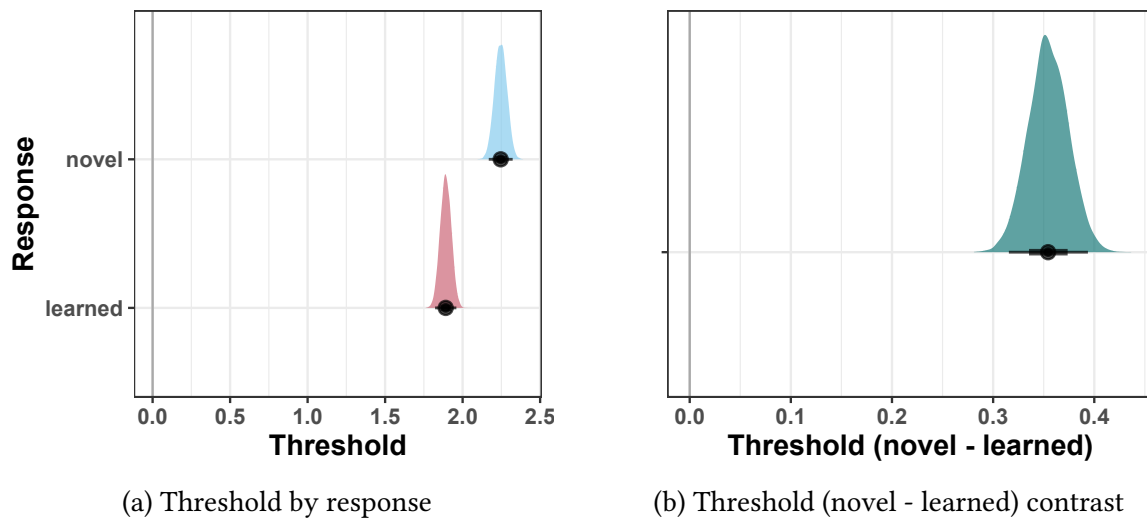


Figure 4. Threshold estimates

Drift rate: Inspection of the posterior distributions shows that there is a familiarity effect in drift rate difference. The quality of the evidence accumulation was greater for Novel faces compared to Learned faces.

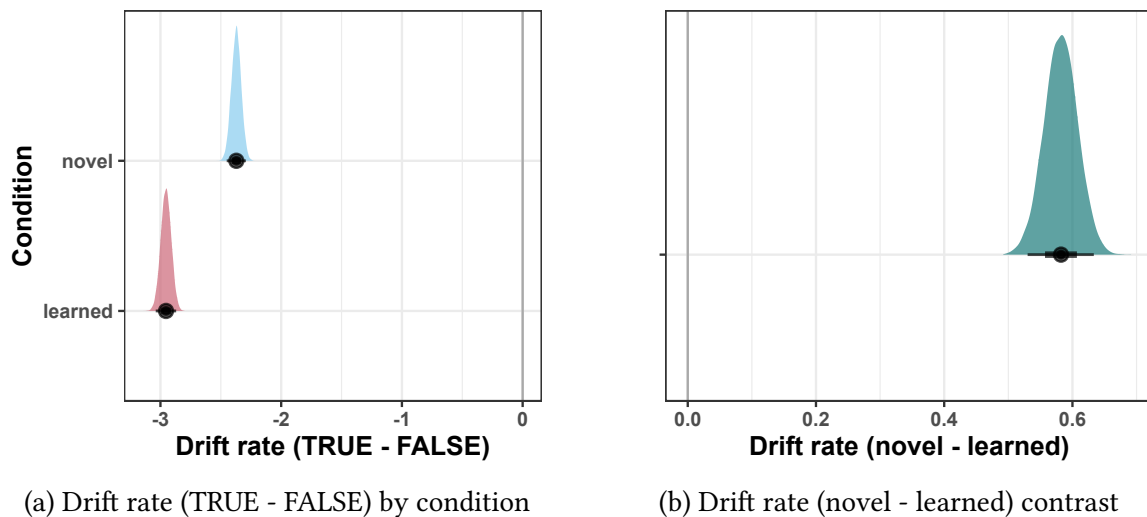


Figure 5. Drift rate estimates

Table 3. Posterior median and 95% quantile intervals for threshold and drift rate parameters.

Parameter	Condition	Median	95% QI
Threshold	novel	2.25	[2.169, 2.323]
Threshold	learned	1.89	[1.822, 1.96]
Threshold contrast	novel - learned	0.35	[0.316, 0.394]
Drift rate	novel	-2.37	[-2.451, -2.293]
Drift rate	learned	-2.95	[-3.039, -2.87]
Drift rate contrast	novel - learned	0.58	[0.53, 0.633]

QI = quantile interval.

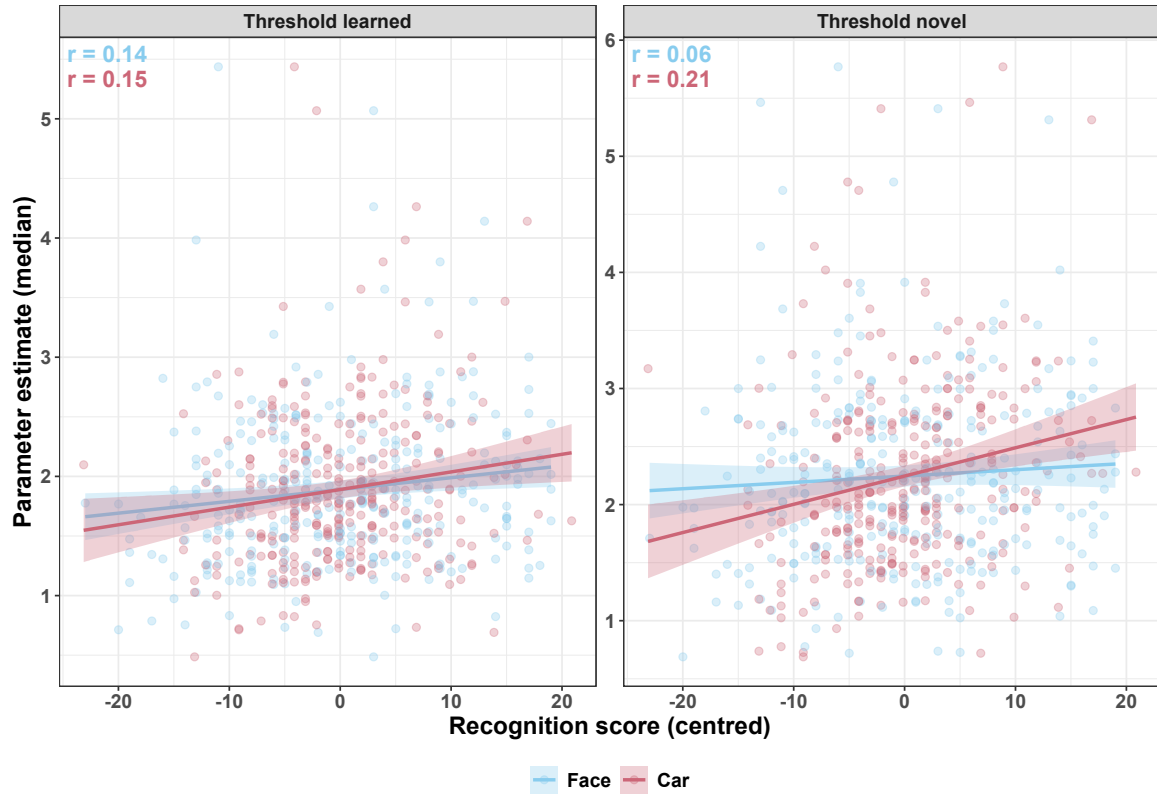
3.2. Correlations

To estimate the association between LBA parameters and recognition abilities (face and car), we computed the linear relationship between the median of the posterior distribution of each estimated parameters of interest and the two recognition abilities. These relationships can be seen in Figure 6 for both recognition ability.

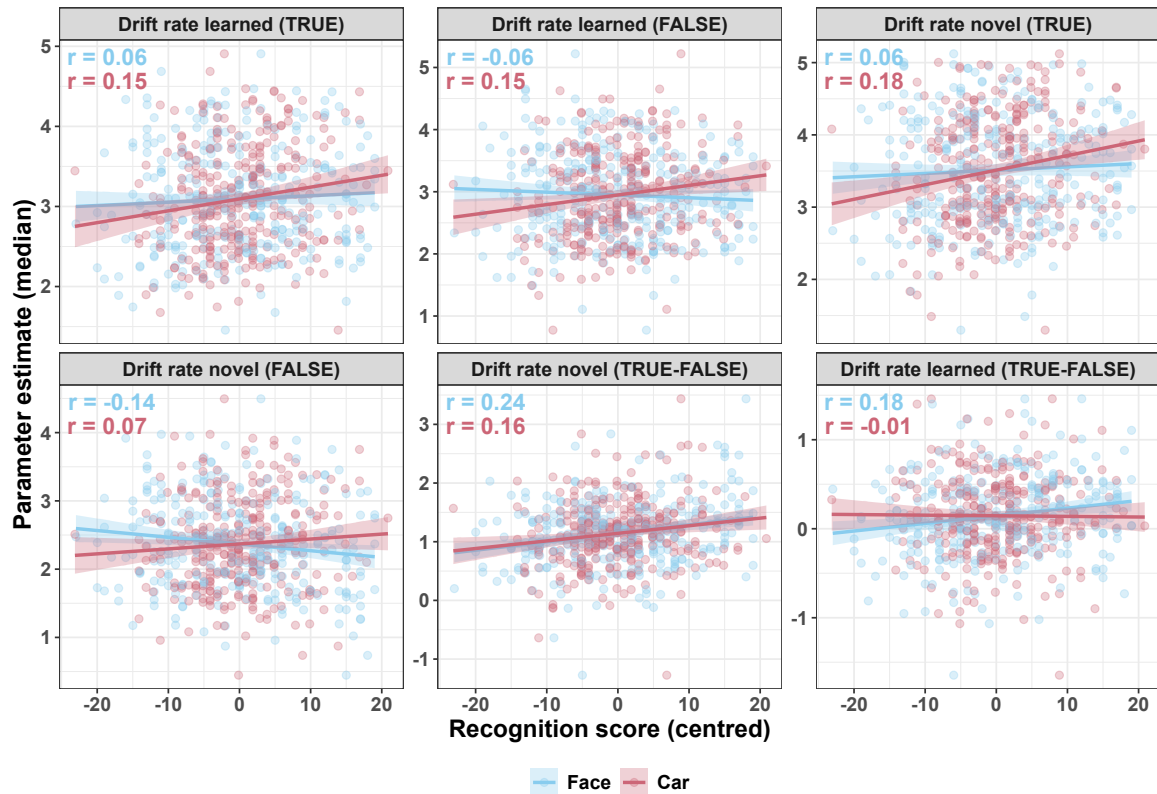
Threshold: Visual inspection of the point estimates correlations shows that when face or car recognition increases, the threshold for learned faces, reflecting the response caution towards the “learned” response increases. The response caution towards novel faces only seems to increase when car recognition ability increases.

Drift rate: Visual inspection of the point estimates correlations shows that when face recognition ability increases, the drift rate (TRUE-FALSE) for learned faces, reflecting the signal-to-noise ratio provided by learned faces, increases as well. The signal-to-noise ratio provided by novel faces increases when either face or car recognition ability increases.

However the strength of these relationship remains weak (the highest r value is 0.24) and we used a Bayesian plausible value correlational approach to determine the uncertainty linked to these associations.



(a) Median threshold correlations with recognition ability



(b) Median drift rate correlations with recognition ability

Figure 6. Point estimates correlations with recognition scores. In (b), drift rate (TRUE) reflects the accumulation of evidence when the response is correct and drift rate (FALSE) the accumulation of evidence when the response is incorrect. Drift rate (TRUE-FALSE) represent the signal-to-noise ratio provided by the stimulus.

3.3. Bayesian plausible value correlational analysis

To estimate the uncertainty linked to the relationships between LBA estimated parameters and face recognition ability, we used a Bayesian plausible value correlational analysis. Like described above (see Methods), instead of using a single point estimate to assess the strength of the relationship between our variables of interest, the posterior distribution of the LBA parameters allows us to use the entire sampling distribution to do it. We calculated the Person’s r between each sample value from our posterior distribution for each parameter of each subject and their face recognition ability. The posterior distributions of threshold plausible correlation values and their contrast are shown in Figure 7. Posterior correlations with true and false drift rate for both conditions and recognition abilities can be found in Figure 8, and correlations with drift rate differences (true-false) by conditions as well as drift rate contrast (novel-false) in Figure 9. For both threshold and drift rate, we evaluated the 95% CI of the distributions. If the 95% CI of the distribution does not include 0, we would be confident in the existence of a relationship between these parameters and the associated recognition ability. Table 4 provides a summary of these distributions.

Threshold: There is a positive relationship between response caution for both learned and novel faces and recognition ability. People with a higher face recognition ability have a higher response caution for learned faces whereas people with a higher car recognition ability have a higher response caution for both learned and novel faces.

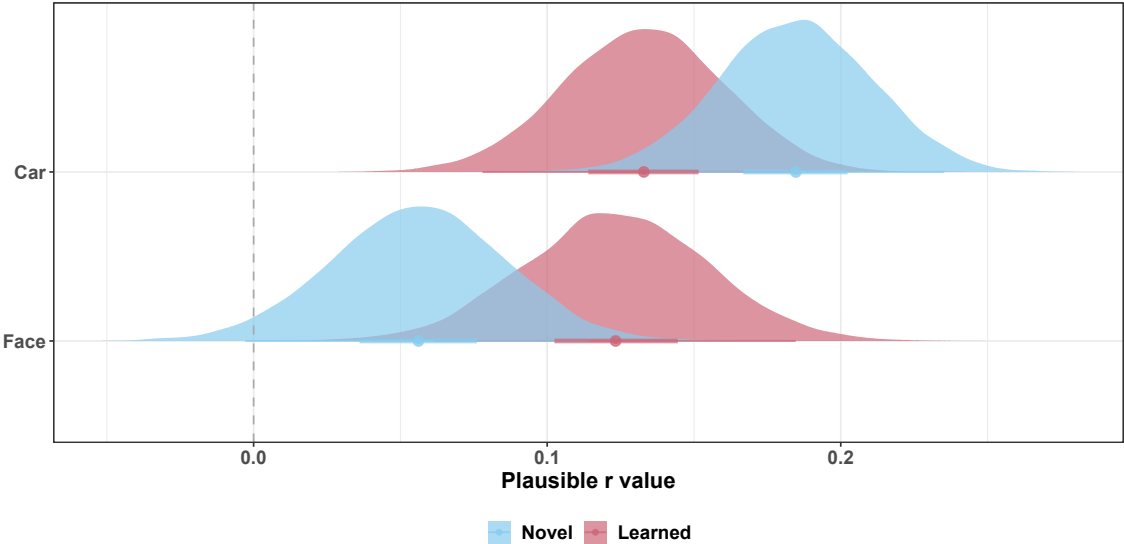


Figure 7. Plausible correlation values distributions between threshold by response and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.

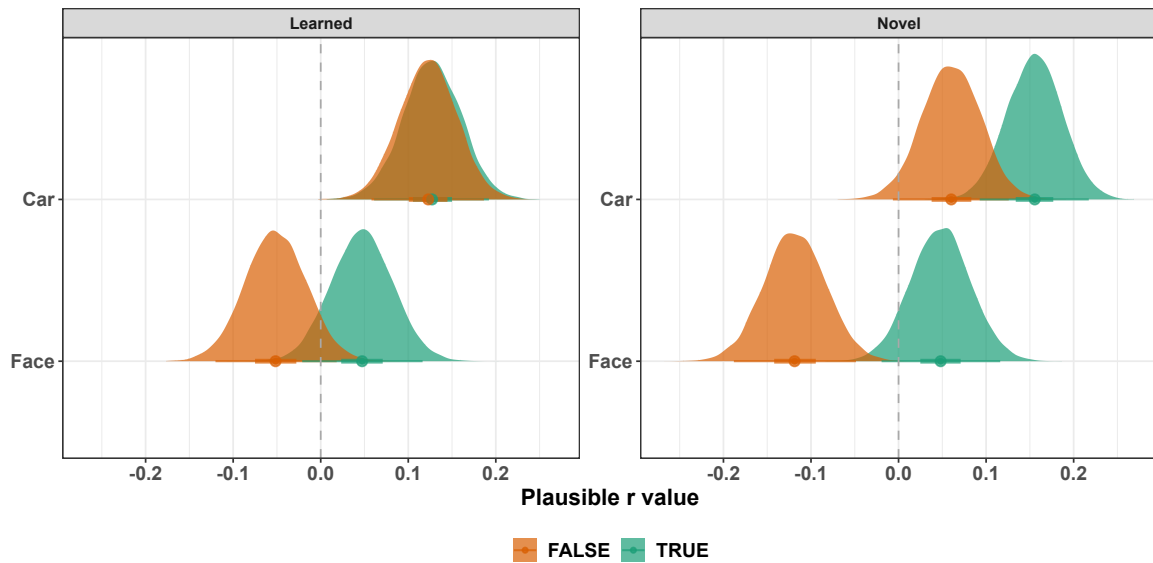


Figure 8. Plausible correlation values distributions between drift rate TRUE and FALSE by stimulus and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.

Drift rate: Regarding drift rate, face recognition ability is positively associated with the signal-to-noise ratio (drift rate TRUE-FALSE) provided by both novel and learned faces. Car recognition ability is only associated with the the signal-to-noise ratio (drift rate TRUE-FALSE) provided by novel faces.

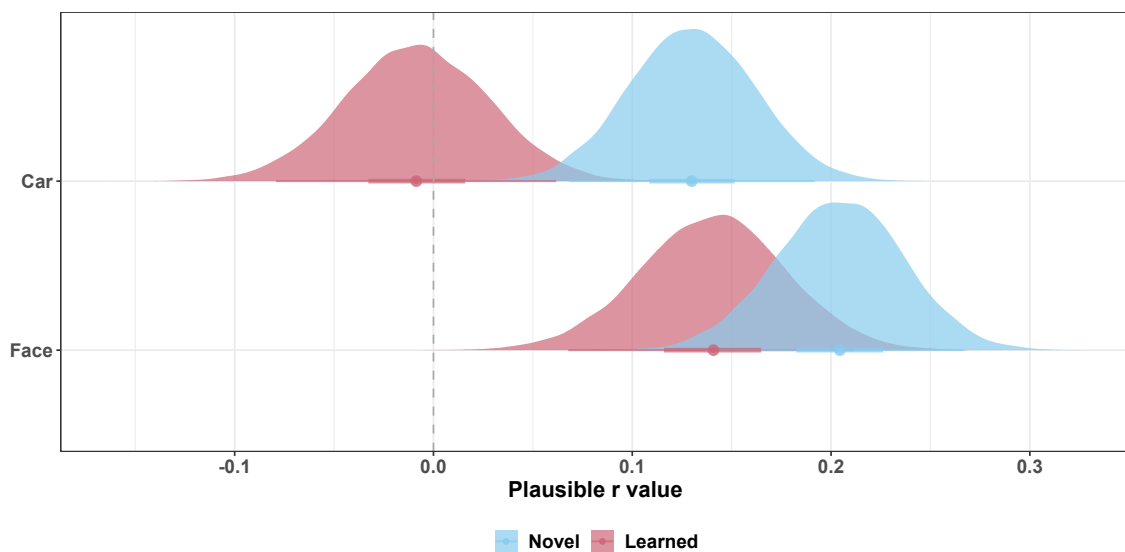


Figure 9. Plausible correlation values distributions between drift rate by stimulus (TRUE-FALSE) and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.

Overall these posterior plausible value correlations show similar observations than the point estimates correlations. However, by taking into account the uncertainty linked to the estimation of the LBA parameters, the confidence in the existence of these associations is increased.

Table 4. Median r and 95% quantile intervals of the plausible value correlations between LBA parameters and recognition ability scores.

Parameter	Condition	Accuracy	Face		Car	
			median r_{Face}	95% QI_Face	median r_{Car}	95% QI_Car
Threshold	Learned	-	0.12	[0.06, 0.18]	0.13	[0.08, 0.19]
Threshold	Novel	-	0.06	[-0.00, 0.11]	0.18	[0.13, 0.24]
Threshold	NA	-	-0.06	[-0.11, -0.02]	0.10	[0.05, 0.14]
Drift	Learned	-	0.14	[0.07, 0.21]	-0.01	[-0.08, 0.06]
Drift	Learned	NA	-0.05	[-0.12, 0.02]	0.12	[0.06, 0.19]
Drift	Learned	NA	0.05	[-0.02, 0.12]	0.13	[0.06, 0.19]
Drift	Novel	-	0.20	[0.14, 0.27]	0.13	[0.07, 0.19]
Drift	Novel	NA	-0.12	[-0.19, -0.05]	0.06	[-0.01, 0.13]
Drift	Novel	NA	0.05	[-0.02, 0.12]	0.16	[0.09, 0.22]
Drift	NA	-	0.05	[-0.02, 0.12]	0.08	[0.01, 0.15]

Note. Blue = 95% QI entirely above zero; red = 95% QI entirely below zero.

3.4. Linear regression

Following our preregistration, the predictions of the two regression models for drift rate can be seen in Figure 10. From visual inspection, the accumulation of signal coming from the stimulus (drift rate TRUE) seems to increase when face recognition ability increases whereas the accumulation of noise (drift rate FALSE) decreases as a function of face recognition ability for both Novel and Learned faces. This pattern of prediction remains the same when accounting for car recognition ability. Inspection of the posterior predictions in Table 5 shows that some of these predicted intervals overlap with 0, and that caution is needed when extrapolating these results.

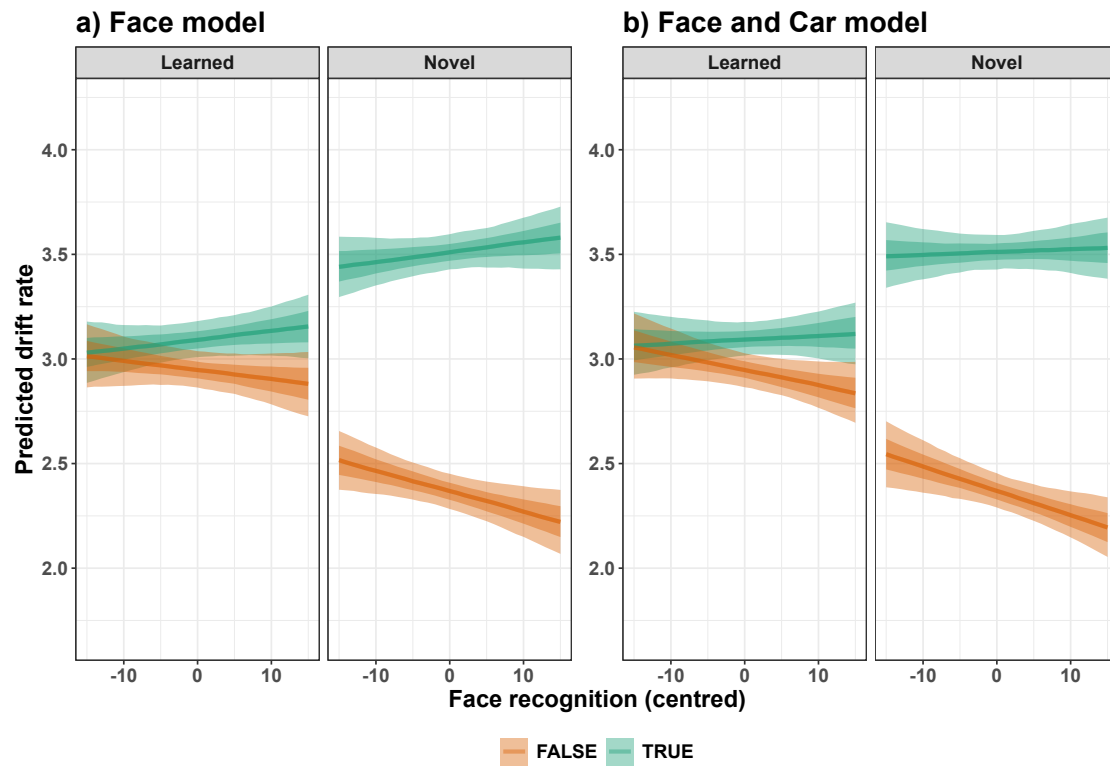


Figure 10. Predicted drift rate values (TRUE and FALSE) from face recognition ability scores for a) Face model and b) Face and Car model

Table 5. Estimates and 95% quantile intervals of the predicted drift rate values slope (TRUE and FALSE) from face recognition ability scores

Parameter	Model	Condition	Accuracy	Estimate	l-95% CI	u-95% CI
Drift slope	Face	Learned	FALSE	-0.0044	-0.0126	0.0039
Drift slope	Face	Novel	FALSE	-0.0097	-0.0179	-0.0014
Drift slope	Face	Learned	TRUE	0.0042	-0.0040	0.0124
Drift slope	Face	Novel	TRUE	0.0047	-0.0035	0.0129
Drift slope	Face + Car	Learned	FALSE	-0.0076	-0.0161	0.0005
Drift slope	Face + Car	Novel	FALSE	-0.0117	-0.0201	-0.0035
Drift slope	Face + Car	Learned	TRUE	0.0017	-0.0068	0.0100
Drift slope	Face + Car	Novel	TRUE	0.0012	-0.0073	0.0093

In Figure 11, the predicted signal-to-noise ratio (drift rate TRUE-FALSE) for both Novel and Learned faces increases as a function of face recognition ability, even when accounting for car recognition ability. As shown in Table 6, these predicted intervals do not overlap with zero. In summary, as people get better at recognizing faces, their ability to accumulate signal and ignoring noisy information in order to recognize Learned faces and to reject Novel faces increases.

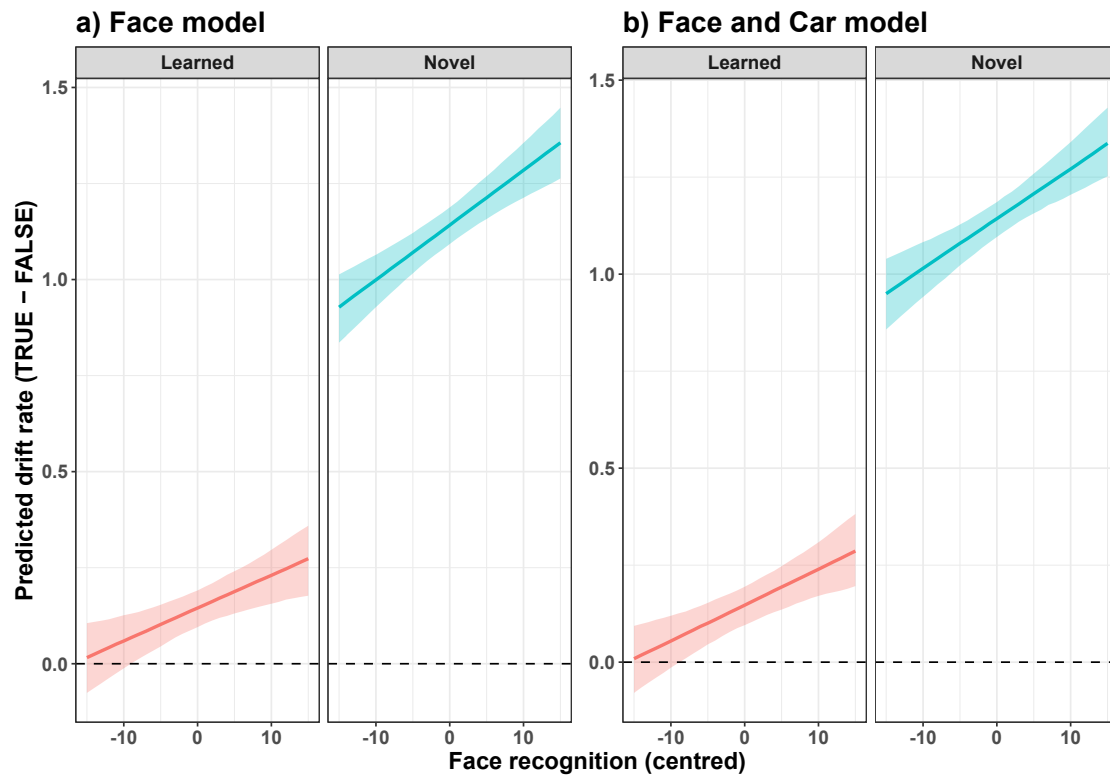


Figure 11. Predicted drift rate values (TRUE-FALSE) from face recognition ability scores for a) Face model and b) Face and Car model

Table 6. Estimates and 95% quantile intervals of the predicted drift rate values slope (TRUE-FALSE) from face recognition ability scores

Parameter	Model	Condition	Estimate	l-95% CI	u-95% CI
Drift slope (TRUE-FALSE)	Face	Learned	0.0086	0.0036	0.0135
Drift slope (TRUE-FALSE)	Face	Novel	0.0144	0.0094	0.0194
Drift slope (TRUE-FALSE)	Face + Car	Learned	0.0092	0.0041	0.0144
Drift slope (TRUE-FALSE)	Face + Car	Novel	0.0129	0.0078	0.0179

4. Discussion

This project combined approaches from experimental, differential and mathematical psychology to investigate individual differences in the computational processes that support face recognition. We show that face recognition ability is positively associated with the rate of evidence accumulation (drift rate parameter) for both learned and novel faces, as well as the response caution (threshold parameter) for learned faces. Further analyses showed that this pattern of results does not appear to reflect a more general object recognition ability, and thus the observed relationship may be partially specific to processing faces. More generally, these findings extend past work on face perception by showing that there is value to be gained by estimating how expertise in face recognition shapes the latent computational processes that support a key dimension of everyday social interactions.

Variation in face recognition ability had a multifaceted impact on parameters in our decision-making model, such that drift rate and threshold were both positively associated with face recognition ability. The drift rate results suggest that people who are generally better at recognizing faces accumulates more recognition signal and are more capable to ignore noisy information coming from the stimulus. Our findings are, therefore, in accordance with previous research that showed that expertise effects are primarily due to a better information extraction (Litchfield &

Donovan, 2017; Thompson et al., 2013; Van Der Gijp et al., 2014). We extend these findings by showing that these effects are not limited to differences between smaller subgroups of experts and novices at the edge of the face recognition distributions, but they concern variation in the general population as well.

Regarding response caution, previous studies have shown that confidence is itself a decision variable used to regulate future behavior (Fleming et al., 2018; Folke et al., 2016). The positive association between threshold and face recognition ability could reflect a better estimation of the task difficulty for people with higher face recognition ability, who voluntarily increase their response caution to ensure a better performance given the task demands. However, future research is required to confirm this suggestion. Overall, our results suggest that not only does high face recognition ability predict a better ability to extract meaningful information for the recognition success but also predict the voluntary strategic decision of the subject to be more cautious.

Moving beyond manifest / observed measures to uncover novel insight into the structure of cognition and expertise in the domain of face perception. Most differential studies in face processing have used test scores to estimate different abilities underlying face processing and investigated if these abilities were related to each other or to other predictors of interest (McCaffery et al., 2018; Noyes et al., 2018). Here we move this research field into a new direction, which feel has many benefits for understanding the cognitive processes involved in social perception and cognition more generally. How the different parameters of the evidence accumulation models are related to underlying psychological constructs is already well established (Evans & Wagenmakers, 2019). One benefit for differential studies is that hypotheses about individual differences in social information processing becomes more specific and explicit (which parameter/psychological construct) and goes beyond performance in term of manifest measures (i.e. accuracy and response times). Another benefit is using a model comparison approach. By comparing the best fitting model across individuals, it becomes possible to assess which participants display a specific effect, as well as as which psychological constructs of the decision-making process best describe different patterns of performance between participants (Alistair et al., 2024; Evans & Wagenmakers, 2019). These benefits could provide meaningful insights regarding how the structure of social cognition varies across individuals.

Although our primary aim was to investigate how face recognition ability predicts computational processes involved in face perception, we conducted exploratory analyses using car recognition ability as a comparison. Car recognition ability positively predicted drift rate for novel faces but not for learned faces, and unlike face recognition ability, it did not predict the discrimination between TRUE and FALSE drift rates for learned faces. These findings suggest that car and face recognition abilities relate differently to the computational processes underlying face recognition. We speculate that this pattern may reflect differences in information processing: object recognition relies primarily on featural processing, whereas face recognition increasingly integrates configural processing as faces become familiar (Cabeza & Kato, 2000; Madole & Cohen, 1995; Schwaninger et al., 2002; Tversky & Hemenway, 1984). Featural processing alone may support evidence accumulation for novel face recognition while also introducing noise, whereas combining featural and configural processing may enhance meaningful evidence accumulation and reduce detrimental noise. Future research should test whether these distinct associations reflect differences in underlying processing mechanisms.

Previous evidence accumulation model development studies have highlighted how the number of trials per cell of the design influences the validity and precision of how model parameters are estimated (Alexandrowicz & Gula, 2020; Lerche et al., 2017; Wagenmakers et al., 2007). Our design (90 trials per condition) and control of error rates in our face recognition task followed the minimum recommendations to apply EAM (Lüken et al., 2025). Nonetheless, adding more trials per condition (e.g., 200 per condition; Boag et al. (2025)) would have increased the robustness of our parameter estimates. Overall the effect sizes of the drift rate predictions could be considered small. However, these effects are consistent with previous literature findings on expertise effects (Litchfield & Donovan, 2017; Thompson et al., 2013; Van Der Gijp et al., 2014), which support our inferences of these effects. If our predictions would have shown the opposite of the expected

expertise effects (drift rate (TRUE - FALSE) decreasing as face recognition ability increases), then we would have emphasized more caution to interpret our small effect size results.

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